From forms to FHIR

Definition-based extraction in Belgian healthcare

Test atelier proposal — FHIR-a-thon, 2–3 June 2026

Annabel Dompas (UZ Leuven) · Axel Vanraes (Tiro.health)

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The problem and the FHIR solution

THE PROBLEM

- Domain experts define forms (checklists, CRFs, synoptic reports) but software vendors build rendering logic from scratch. **No shared specification.**
- Meaningful clinical data is trapped in forms. Extracting structured, coded data is tedious with **no shared semantics.**

THE FHIR SOLUTION

- Forms modelled as **FHIR Questionnaire**, responses as **QuestionnaireResponse**. Decouples domain expert from vendor.
- SDC **$extract** transforms responses into discrete FHIR resources (Observations, Conditions, Procedures) — structured, coded, and FAIR.

SDC definition-based extraction workflow

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USE CASE 1

Nurse checklists (thuishospitalisatie)

- OPAT (IV antibiotics at home) & antitumoral therapy forms
- Prepared by UZ Leuven - maps to the FOD-funded thuishospitalisatie pilot (since March 2026)

USE CASE 2

Mandatory registry population

- Registry forms (QERMID, cancer registration) extracted to registry-specific data models
- Relevant for Sciensano/HD4DP, BCR, and BSP

CONFIRMED PARTICIPANTS

UZ Leuven · Tiro.health · Amaron · Axians (BCR) · nexuzhealth

Shared test infrastructure: Tiro.health public $extract API

No participant needs to build extraction logic

3-role architecture diagram (registry owner → data provider → data transport) available on request

Questions?

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Co-organisers

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Axel Vanraes — Tiro.health

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